FOLLICA: Scalp Trichoscopy & Biomedical Decision Support

Biomedical System Design Flow & Cyber-Physical Instrumentation Architecture

System Classification: Software as a Medical Device (SaMD) / Dual-Pipeline Computer Vision & Triage Engine
Architectural Foundation: Standard Biomedical Instrumentation Design Flow (Transducer to Clinical Diagnosis)
Clinical Benchmarks: Dichromatic Reflection Model | L'Oreal Dandruff Atlas (PMC11975186) | Traya 3-Science Axis

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1. Executive Summary & Biomedical Problem Formulation

Follica is an open-source, non-invasive digital scalp trichoscopy and systemic triage system designed to bridge the gap between high-cost clinical optical instrumentation (such as polarized FotoFinder dermoscopes) and non-standardized commercial e-commerce intake funnels (such as Traya). The system operates on a dual-pipeline architecture: deterministic classical computer vision micro-feature extraction running parallel with deep learning feature representations, fused with a systemic multi-factor intake model covering gastrointestinal absorption, metabolic rate, and endocrine profiles.

2. Complete Mapping to the Biomedical Instrumentation Design Flow

The standard Biomedical Design Flow outlines the transformation of an organic clinical pathology into a digitized, processed, and clinically actionable diagnostic output. Below is the direct, stage-by-stage engineering mapping for Follica:

Design Flow Stage	Biomedical Engineering Implementation in Follica	Signal & Algorithmic Details
1. Clinical Condition	Scalp dermatological and follicular pathologies including Seborrheic Dermatitis, Alopecia Areata, Androgenetic Thinning, Psoriasis, and Normal Scalp Homeostasis.	Includes systemic etiologies: acute telogen effluvium, gut-mediated malabsorption, and thyroid dysfunction.
2. Physiological Parameters	Surface optical reflection, stratum corneum desquamation, follicular spatial distribution, cutaneous microvascular perfusion, and lipid barrier gloss.	Quantified into 4 bounded sub-scores (0-25 each): Flake, Redness/Erythema, Sebum/Lipid, and Follicular Density.
3. Mode of Measurement	Non-invasive Optical Trichoscopy in the visible electromagnetic spectrum (wavelengths 400 nm to 700 nm).	Multi-angle macroscopic photographic capture (Crown, Hairline, Parting, Temples).
4. Transducer Design	Smartphone solid-state CMOS/CCD photodiode sensor array with Bayer Color Filter Array (RGGB pattern).	Converts incident photons to photo-generated electron charge packets, then to proportional analog voltages.
5. Calibration Signal	Optical: Gray-World White Balance algorithm normalizing ambient room illuminants to D65 standard. Clinical: 6-point exfoliating severity scale calibrated per Flament et al. 2025 (PMC11975186, r^2 = 0.952).	Ensures colorimetric invariance against warm yellow (2700K) or cool fluorescent (6500K) room lighting.
6. Filter & Pre-Amplifier Design (Noise Suppression)	Software Pre-Processing Quality Gate & Optical Filtering: - Dichromatic Specular Glare Mask (Shafer Model) suppressing flash reflection artifacts. - Discrete 2D Laplacian operator filtering out motion/defocus blur (var < 60.0). - Luminance histogram boundary gate (35 < mean < 225) rejecting under/overexposure. - EXIF optical parser detecting high sensor grain (ISO > 800) and slow shutter speed.	Eliminates flash reflections being misclassified as white dandruff flakes, and prevents motion-blurred images from clouding diagnosis.

7. Signal Digitization (ADC)	Hardware Analog-to-Digital Converter integrated on the camera image signal processor (ISP). - Spatial Sampling: Standardized 512 x 512 discrete spatial grid (262,144 pixels). - Quantization Depth: 24-bit Truecolor (8 bits / 256 discrete levels per channel).	Generates normalized numerical tensor I(x, y) in RGB color space with bounded dynamic range [0, 255].
8. Digital Signal Processing (DSP)	Dual Classical Computer Vision + Deep Learning Engine: - Morphological Top-Hat filtering isolating high-frequency keratin squames. - Local Laplacian Texture Variance Discriminator (var >= 8.0) separating rough flakes from smooth glare fringes. - MobileNetV4 / EfficientNet-B0 deep convolutional feature extraction. - ORB Keypoint Homography Registration for serial longitudinal delta tracking (Day 0 vs Day 14/30). - Traya 3-Science Multi-Factor Correlation Engine (Gut, Energy, Hormones).	Replaces opaque black-box deep learning with explainable, deterministic mathematical metrics combined with systemic clinical correlation.
9. Wireless Transmission	Encrypted client-server payload transmission over TLS (HTTPS). Multipart/form-data schema transmitting image bytes and structured JSON questionnaire context to FastAPI.	AES-256 encrypted at rest in database storage with user-selectable 30-day automated purge window.
10. Data Display & Storage	Client-side Single Page Application (SPA) displaying composite Scalp Health Score (0-100 SHS), 4 discrete sub-score tracks, live risk indicator, and longitudinal comparison.	Secure, pseudonymous session storage preventing biometric photo linkage to personally identifiable information (PII).
11. Clinical Diagnosis & Triage Support	Server-Side 3-Tier Clinical Triage Engine: - Level 1: Minor Home Routine (Preventative barrier hydration). - Level 2: Monitored Routine (Active ingredients with 14-day tracking; strict server confidence gate < 0.65). - Level 3: Escalation (Automated generation of 1-Page Clinical Summary PDF pass for certified dermatologist review).	Follows medical ethics standards by operating as decision support rather than unlicensed medical diagnosis.

3. Scientific Innovations & Core Algorithmic Proofs

A. Flash Glare Suppression via the Shafer Dichromatic Reflection Model

A critical challenge in mobile scalp trichoscopy is that flash or ring light creates intense, circular specular hotspots on the curved scalp surface. In standard morphology algorithms, these bright white highlights are falsely identified as white dandruff flakes (causing healthy scalps to be diagnosed with Seborrheic Dermatitis).
Follica resolves this physically via the Shafer Dichromatic Reflection Model: Total reflectance L(x) = L_diffuse(x) + L_specular(x). Flash reflection is purely specular and achromatic (V > 230, |R-G| + |G-B| < 25, S < 35 in HSV). Follica constructs a binary specular mask, suppresses these pixels from the luminance channel prior to Top-Hat morphology, and routes the specular energy exclusively to the sebum/gloss sub-score where surface shine belongs. Remaining candidates are evaluated through a local Laplacian texture variance discriminator; genuine keratin squames have rough irregular boundaries (var >= 8.0), whereas residual specular fringes have smooth falloff and are rejected.

B. Clinical Dandruff Atlas Calibration (PMC11975186 Benchmark)

Rather than using an arbitrary threshold, Follica's flake scoring algorithm is calibrated directly against the clinical dandruff grading atlas published by Flament et al. (2025, PMC11975186 / Journal of Cosmetic Dermatology), which established r^2 = 0.952 correlation with 15 expert dermatologists. The system maps measured particle surface coverage to the validated 6-point scale: Grade 0 (<0.1% coverage -> 25.0 score), Grade 1 (0.1-0.5% -> 22.0-24.9), Grade 2 (0.5-1.2% -> 18.0-21.9), Grade 3 (1.2-2.5% -> 13.0-17.9), Grade 4 (2.5-4.0% -> 7.0-12.9), and Grade 5 (>4.0% -> 0.0-6.9).

C. Systemic Biomarker Integration (Traya 3-Science Model)

Clinical trichology demonstrates that hair loss cannot be diagnosed accurately by superficial scalp appearance alone. Diffuse thinning and sudden shedding (Telogen Effluvium) are driven by systemic physiology. Following the Traya 3-Science diagnostic framework (Dermatology + Nutrition + Internal Metabolism), Follica integrates self-reported systemic biomarkers into its clinical correlation engine:

1. **Gastrointestinal Axis:** Identifies bloating, chronic constipation, or IBS as malabsorption risks that starve hair matrix keratinocytes of iron, zinc, and biotin.
2. **Metabolic Axis:** Flags chronic sluggish fatigue and rapid weight fluctuations as systemic metabolic load indicators.
3. **Endocrine Axis:** Correlates thyroid dysfunction and PCOS with follicular cycle elongation and local DHT binding affinity, triggering automated blood panel recommendations (TSH, Free T3/T4, Ferritin, Vitamin D3).

4. Automated Quality Assurance & Verification Suite

The complete Follica codebase is backed by an automated verification suite consisting of **54 end-to-end tests** executed via pytest. The test suite covers: dual-pipeline sub-score invariants, server-side confidence gate rerouting (<0.65 threshold), pre-inference Laplacian blur and overexposure rejections, adversarial geometries (extreme aspect ratios, sensor noise, corrupted bytes), longitudinal ORB registration, flash glare suppression on synthetic healthy scalps, and metabolic/gut health correlation synthesis. All 54 tests pass with 100% success rate.

5. Academic Conclusion

By adhering strictly to the Biomedical Instrumentation Design Flow, Follica demonstrates how smartphone sensors can be transformed into a clinically grounded diagnostic screening instrument. The project achieves technical parity with commercial systems like Traya on systemic anamnesis, while outperforming them on the computer vision front through deterministic specular glare separation and open-source scientific transparency.